

electromagnetic energy, or atomic or subatomic particles in the targeted direction to cause intended damage to the enemy's equipment, facilities and personnel. Intended damage could be lethal or non-lethal.

Ever since H.G. Wells published *War of the Worlds* in 1898, DEWs have been a recurring theme in science fiction literature. Idea of a death ray, which can instantly destroy or burn a target at a distance, in fact, dates back to a belief that Archimedes used a burning glass to set afire Roman ships during the siege of Syracuse in 212 B.C. Although many images of the death ray depict Archimedes with a parabolic mirror, use of a set of individual flat mirrors appropriately positioned seemed to be a more practical implementation of the concept (Fig. 1).

Though the story has long been dismissed as a myth, interest generated by it has led to a number of experiments being conducted to verify the technical feasibility of such an event. Experiments conducted by Comte de Buffon and Dr Ioannis Sakkas, and more recently by students of Massachusetts Institute of Technology (MIT), USA, have established the feasibility of such an occurrence.

Buffon assembled 168 mirrors, 20.3cm x 25.4cm (8-inch x 10 inch) each, adjusted to produce the smallest image 45.7m (150-feet) away. The array turned out to be a formidable weapon. With this elaborate setup, he performed several experiments. He demonstrated igniting a creosoted plank at 20.1m (66-feet) distance using only 40 mirrors. 128 mirrors could ignite a pine plank instantly and, in another experiment, 45 mirrors melted 2.7kg (six pounds) of tin at 6.1m (20-feet).

In another effort, Dr Sakkas lined up nearly 60 Greek sailors,



Fig. 2: Technical feasibility of death ray, an MIT experiment

each holding an oblong mirror tipped to catch the Sun's rays and directed these at a wooden ship 48.7m (160-feet) away. The ship caught fire at once.

As recently as 2009, MIT students carried out an experiment with 11.7sqm (127-sqft) mirrors focusing solar radiation on to a boat 30.5m (100-feet) away, causing a sustained flame and confirming technical feasibility of what Archimedes might have achieved with his death ray (Fig. 2).

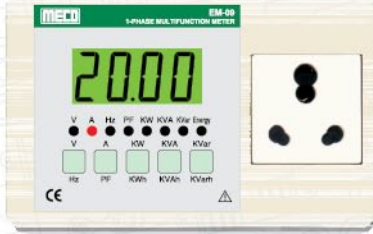
DEWs versus KEWs

At the most fundamental level, DEWs share the concept of delivering a large amount of stored energy from the weapon to the target to produce structural and incendiary damage effects. Kinetic energy weapons (KEWs) deliver this effect at subsonic or supersonic speeds while DEWs do so at the speed of light.

Both KEWs and DEWs need to address two fundamental issues. The first major concern is related to travel or propagation through the atmosphere and hitting the target. In the case of KEWs, it is getting the projectile to successfully travel through the atmosphere and hit the target. In the case of DEWs, it is the propagation of high-energy beams such as high-power electromagnetic radiation or high-energy particle beams through the atmosphere and



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